

BHARGOB DEKA

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SUMMARY

Senior AI/ML developer with 5+ years of experience building forecasting, anomaly detection, and vertical AI agent systems. At Hydro-Quebec, I developed forecasting and anomaly detection models that improved predictive accuracy and reduced false alarms. Currently, I lead the development of AI Agents and demand forecasting projects at Jesta I.S., managing cross-functional teams and deploying scalable cloud solutions with AWS and Azure. Published author with hands-on expertise in LLMs, RAG, and technical writing.

SKILLS

Languages: Python, SQL, Javascript, MATLAB, C++

Frameworks: TensorFlow, PyTorch, Kedro, MLFlow, FastAPI, Scikit-learn, React

Generative AI: LangChain, LangGraph, LangSmith, CrewAI, Hugging Face, OpenAI, Ollama

Cloud & DevOps: AWS (Batch, S3, Lambda, Sagemaker), Azure pipelines, Docker, Terraform, GitHub Actions

WORK EXPERIENCE

Senior AI-ML Developer

Jesta I.S.

Oct 2024 – present

Montreal, QC

- Led the design and development of the company's first AI Agent, enabling clients to obtain business insights from natural language queries against their own data. Prototyped with Gradio, then built a scalable backend with FastAPI and a modern frontend using React and TypeScript. Deployed the solution via Azure DevOps pipelines, and using Langsmith for experiment tracking and evaluation.
- Engineered an end-to-end demand forecasting system by developing separate LightGBM regression models for short-term and pre-season forecasts, and automated the model lifecycle using Kedro for pipeline orchestration, AWS Batch for job execution, and MLflow for experiment tracking and evaluation.
- Developed AutoEncoder and Random Forest models for a fraud detection POC, enhancing detection accuracy and demonstrating feasibility.

ML Researcher

Hydro-Quebec

Jan 2023 – May 2024

Montreal, QC

- Developed probabilistic LSTM models for forecasting using data from 400+ dams across Quebec, supporting data-driven structural health monitoring.
- Combined it with a real-time anomaly detection framework using switching Kalman filter, reducing false alarms by almost 50%.
- Built an end-to-end PyTorch framework for training, evaluating, and monitoring ML models, integrated with MLFlow on AWS.
- Improved predictive accuracy by 10% by modeling heteroscedastic uncertainty in time series data.
- Partnered with HQ business stakeholders to deliver hands-on training and support, ensuring effective adoption of ML solutions.

Graduate Research Assistant

Polytechnique Montreal, Funding Partner: Hydro-Quebec

Sep 2018 – Dec 2022

Montreal, Quebec

- Developed a multiplicative state-space model to accurately capture non-harmonic periodic patterns in structural health monitoring applications.
- Introduced AGVI, a novel approach for real-time uncertainty quantification in Bayesian machine learning models, eliminating the need for hyperparameter tuning and reducing training time by over an order of magnitude.

Graduate Research Assistant

McGill University, Funding Partner: Environment Canada

Sep 2015 – Mar 2018

Montreal, Quebec

- Designed and implemented an enhanced statistical downscaling methodology by integrating multiple techniques to project future climate scenarios.
- Evaluated and optimized statistical models for downscaling daily and hourly climate variables using clustering, PCA, stepwise regression, and quantile-quantile mapping for bias correction.

Teaching Assistant

McGill University

Sep 2016 – Apr 2017

Montreal, Quebec

- Led tutorial sessions for two courses-Probabilistic Systems and Civil Engineering System Analysis-for a cohort of 120 undergraduate students in the Civil and Mining Engineering Department.

Structural Designer

NORSAR

Jan 2014 – July 2015

Guwahati, India

- Contributed to the EQRisk Project by developing a seismic risk model for Guwahati city, supporting urban resilience planning.
- Created detailed 2D and 3D finite element models of various building typologies to assess and optimize structural performance.

EDUCATION

Polytechnique Montreal

Ph.D. in Civil Engineering (GPA 4.0/4.3)

Sep 2018 - Dec 2022

Montreal, QC

McGill University

M.Eng. in Civil Engineering (GPA 3.88/4.0)

Sep 2015 - March 2018

Montreal, QC

OPEN SOURCE PROJECTS

RAG Code Assistant for Speckle

Developed an advanced RAG code assistant for querying Speckle's developer documentation. This assistant retrieves relevant documents, grades them, generates responses, and checks for hallucinations using the LangGraph workflow.

Multi-Modal RAG for Visual Question & Answering

Created a multi-modal RAG chatbot using LangChain and GPT-4. Leveraged the Unstructured library for document parsing, a multi-vector retriever for multi-modal RAG, the DeepEval library for evaluation, and Streamlit for the user interface.

Hotel Recommender using a Multi-Agent Framework

Designed a hotel recommender system using CrewAI and LangChain. Developed two AI agents capable of collaborating autonomously to find the perfect listing based on user input criteria.

Fine-Tuning Vision Transformer for Image Classification

Developed a houseplant health classifier by full and QLoRA fine-tuning a vision transformer on a custom houseplant dataset and deployed an interactive Gradio app on Hugging Face Spaces for real-time assessment.

PUBLICATIONS

Analytically Tractable Heteroscedastic Uncertainty Quantification in Bayesian Neural Networks for Regression Tasks

Neurocomputing

Deka, B., Ha Nguyen, L., Goulet, J. A., 2023

Approximate Gaussian Variance Inference for State-Space Models

International Journal of Adaptive Control and Signal Processing

Deka, B. and Goulet, J. A., 2023